TP5- Implement etherchannel

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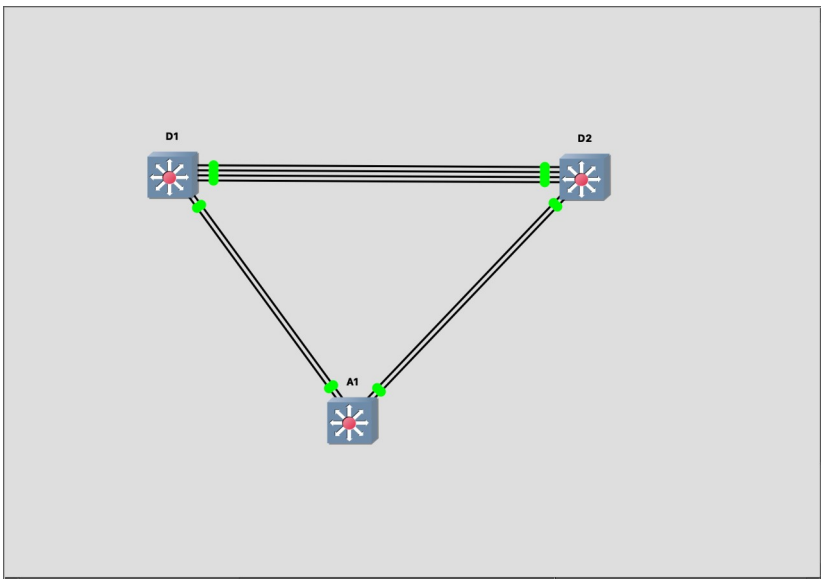
Topology



Bmarco11 — D1 — telnet 192.168.71.128 5000 — 80×24

```
*Oct 27 00:47:33.068: %SYS-5-CONFIG_I: Configured from console by console
D1#show interfaces e1/0 switchport
Name: Et1/0
Switchport: Enabled
Administrative Mode: dynamic auto
Operational Mode: static access
Administrative Trunking Encapsulation: negotiate
Operational Trunking Encapsulation: native
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Administrative Native VLAN tagging: enabled
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk Native VLAN tagging: enabled
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk associations: none
Administrative private-vlan trunk mappings: none
Operational private-vlan: none
Trunking VLANs Enabled: ALL
Pruning VLANs Enabled: 2-1001
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 80×24

```
Protected: false
Appliance trust: none
A1#
A1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
A1(config)#interface e1/0
A1(config-if)#switchport mode dynamic desirable
A1(config-if)#end
A1#
*Oct 27 01:10:50.604: %SYS-5-CONFIG_I: Configured from console by console
A1#show interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Et1/0       desirable        n-isl          trunking      1

Port        Vlans allowed on trunk
Et1/0       1-4094

Port        Vlans allowed and active in management domain
Et1/0       1

Port        Vlans in spanning tree forwarding state and not pruned
Et1/0       1
A1#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 80×24

```
Et1/0       1
A1#
*Oct 27 01:14:54.887: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
1, changed state to down
*Oct 27 01:14:55.890: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
1, changed state to up
A1#show interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Et1/0       desirable        n-isl          trunking      1
Et1/1       auto             n-802.1q       trunking      1

Port        Vlans allowed on trunk
Et1/0       1-4094
Et1/1       1-4094

Port        Vlans allowed and active in management domain
Et1/0       1
Et1/1       1

Port        Vlans in spanning tree forwarding state and not pruned
Et1/0       1
Et1/1       none
A1#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 80×24

```
A1(config-if-range)#exit
*Oct 27 01:22:42.873: %LINK-3-UPDOWN: Interface Ethernet1/0, changed state to up
*Oct 27 01:22:42.873: %LINK-3-UPDOWN: Interface Ethernet1/1, changed state to up
*Oct 27 01:22:43.878: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
0, changed state to up
*Oct 27 01:22:43.879: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
1, changed state to up
A1(config-if-range)#exit
A1(config)#
*Oct 27 01:22:44.037: %SPANTREE-7-RECV_1Q_NON_TRUNK: Received 802.1Q BPDU on non
 trunk Ethernet1/0 VLAN1.
*Oct 27 01:22:44.037: %SPANTREE-7-BLOCK_PORT_TYPE: Blocking Ethernet1/0 on VLAN0
001. Inconsistent port type.
A1(config)#end
A1#
*Oct 27 01:22:50.302: %SYS-5-CONFIG_I: Configured from console by console
A1#show interfaces trunk
A1#show interfaces e1/0 switchport | i Mode
Administrative Mode: dynamic desirable
Operational Mode: static access
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Capture Mode Disabled
A1#
```

Bmarco11 — D2 — telnet 192.168.71.128 5001 — 80×24

```
*Oct 27 01:35:36.390: %LINK-3-UPDOWN: Interface Ethernet1/3, changed state to up
*Oct 27 01:35:37.388: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
0, changed state to up
*Oct 27 01:35:37.397: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
1, changed state to up
*Oct 27 01:35:37.397: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to up
*Oct 27 01:35:37.397: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to up
*Oct 27 01:35:37.397: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
2, changed state to up
*Oct 27 01:35:37.397: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
3, changed state to up
D2(config)#end
D2#
*Oct 27 01:35:43.998: %SYS-5-CONFIG_I: Configured from console by console
D2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
D2(config)#interface range e1/2-3
D2(config-if-range)#switchport nonegotiate
D2(config-if-range)#end
D2#
*Oct 27 01:43:07.452: %SYS-5-CONFIG_I: Configured from console by console
D2#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 80×24

```
3, changed state to up
A1(config)#end
A1#
*Oct 27 01:40:53.160: %SYS-5-CONFIG_I: Configured from console by console
A1#show interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Et1/0       on               802.1q         trunking      1
Et1/1       on               802.1q         trunking      1
Et1/2       on               802.1q         trunking      1
Et1/3       on               802.1q         trunking      1

Port        Vlans allowed on trunk
Et1/0       1-4094
Et1/1       1-4094
Et1/2       1-4094
Et1/3       1-4094

Port        Vlans allowed and active in management domain
Et1/0       1
Et1/1       1
Et1/2       1
Et1/3       1
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 80×24

```
A1#show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)          -        Et1/2(P)    Et1/3(P)
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 80×24

```
1       Po1(SU)          -        Et1/2(P)    Et1/3(P)

A1#show spanning-tree

VLAN0001
  Spanning tree enabled protocol rstp
  Root ID    Priority    32769
             Address     aabb.cc00.0100
             Cost        100
             Port        5 (Ethernet1/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     aabb.cc00.0300
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et1/0               Root FWD 100       128.5    Shr
Et1/1               Altn BLK 100       128.6    Shr
Po1                 Altn BLK 56        128.65   Shr
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 80×24

```
A1(config)#vlan 999
A1(config-vlan)#name native_vlan
A1(config-vlan)#exit
A1(config)#interface port-channel 1
A1(config-if)#switchport trunk native vlan 999
A1(config-if)#e
*Oct 27 01:52:10.037: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discover
ed on Ethernet1/3 (999), with D2 Ethernet1/3 (1).
A1(config-if)#end
A1#
*Oct 27 01:52:12.831: %SYS-5-CONFIG_I: Configured from console by console
A1#show interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Et1/0       on               802.1q         trunking      1
Et1/1       on               802.1q         trunking      1
Po1         on               802.1q         trunking      999

Port        Vlans allowed on trunk
Et1/0       1-4094
Et1/1       1-4094
Po1         1-4094

Port        Vlans allowed and active in management domain
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 80×24

```
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)          -        Et1/2(P)    Et1/3(P)
2      Po2(SD)         PAgP      Et1/0(I)    Et1/1(I)
 --More--
```

Bmarco11 — D1 — telnet 192.168.71.128 5000 — 80×24

```
999  enet  100999     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

Remote SPAN VLANs
-------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------

D1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
D1(config)#vlan 999
D1(config-vlan)#name native_vlan
D1(config-vlan)#exit
D1(config)#interface port-channel 2
D1(config-if)#switchport trunk native vlan 999
D1(config-if)#end
D1#
*Oct 27 01:57:30.410: %SYS-5-CONFIG_I: Configured from console by console
D1#
```

Bmarco11 — D2 — telnet 192.168.71.128 5001 — 80×24

```
        R - Layer3      S - Layer2
        U - in use      N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)          -       Et1/2(P)    Et1/3(P)
3      Po3(SD)        LACP      Et0/0(s)    Et0/1(s)    Et0/2(s)
                                 Et0/3(s)

D2#
D2#
```

Bmarco11 — D1 — telnet 192.168.71.128 5000 — 80×24

```
D1#
*Oct 27 02:02:58.414: %SYS-5-CONFIG_I: Configured from console by console
D1#show interfaces trunk | i Port|Po3
Port        Mode             Encapsulation  Status        Native vlan
PPort        Vlans allowed on trunk
Port        Vlans allowed on trunk
Port        Vlans allowed and active in management domain
Port        Vlans in spanning tree forwarding state and not pruned
D1#show interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Et0/0       on               802.1q         trunking      1
Et0/1       on               802.1q         trunking      1
Et0/2       on               802.1q         trunking      1
Et0/3       on               802.1q         trunking      1
Et1/0       on               802.1q         trunking      1
Et1/1       on               802.1q         trunking      1

Port        Vlans allowed on trunk
Et0/0       1-4094
Et0/1       1-4094
Et0/2       1-4094
Et0/3       1-4094
Et1/0       1-4094
```